

SkillSwap

Database Design Package

CP476B Internet Computing · Spring 2026

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DBMS: MySQL 8+ | **Charset:**
utf8mb4_unicode_ci

Milestone: 02

Schema Overview

The SkillSwap database consists of **8 tables** designed to 3rd Normal Form (3NF). All relationships are enforced via foreign key constraints with appropriate CASCADE rules. Every table includes a surrogate integer primary key. Sensitive data (passwords) are never stored in plaintext — only bcrypt hashes.

Table	Purpose	Key Relationships
users	All platform users (customer / provider / admin)	Root entity — referenced by most tables
categories	Lookup table for service categories	FK from services.category_id
services	Service listings created by providers	FK → users, categories
bookings	Links customers to provider services	FK → services, users (x2)
reviews	One review per completed booking	FK → bookings, users
conversations	Unique thread between two users	FK → users (x2), UNIQUE pair
messages	Individual messages inside a conversation	FK → conversations, users
reports	Moderation reports on services/reviews/users	FK → users (reporter + admin)

TABLE DEFINITIONS

users

Column	Type	Constraints	Notes
user_id	INT	PK, AUTO_INCREMENT	Surrogate key
first_name	VARCHAR(50)	NOT NULL	
last_name	VARCHAR(50)	NOT NULL	
email	VARCHAR(100)	NOT NULL, UNIQUE	Login identifier
password_hash	VARCHAR(255)	NOT NULL	bcrypt hash — never plaintext
role	ENUM	NOT NULL, DEFAULT 'customer'	'customer', 'provider', 'admin'
bio	TEXT	NULL	Optional profile text
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Soft disable by admin
created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	

Note: `uq_users_email` enforces one account per email address.

categories

Column	Type	Constraints	Notes
category_id	INT	PK, AUTO_INCREMENT	Surrogate key
name	VARCHAR(50)	NOT NULL, UNIQUE	e.g. Programming, Tutoring, Design

Note: Pre-seeded with 6 values. Prevents VARCHAR duplication in services table.

services

Column	Type	Constraints	Notes
service_id	INT	PK, AUTO_INCREMENT	Surrogate key
user_id	INT	FK → users.user_id, CASCADE	Provider who owns this listing
category_id	INT	FK → categories.category_id	Normalized category reference
title	VARCHAR(100)	NOT NULL	
description	TEXT	NOT NULL	
price	DECIMAL(10, 2)	NOT NULL, CHECK(price > 0)	Positive price enforced
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Soft-delete
created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	

Note: Indexes on `category_id`, `user_id`, `is_active` for fast marketplace queries.

bookings

Column	Type	Constraints	Notes
booking_id	INT	PK, AUTO_INCREMENT	Surrogate key
service_id	INT	FK → services.service_id, CASCADE	
customer_id	INT	FK → users.user_id	The buyer
provider_id	INT	FK → users.user_id	Denormalized for query speed
booking_status	ENUM	DEFAULT 'Pending'	Pending Active Completed Rejected Cancelled
message	TEXT	NULL	Optional request message
booking_date	TIMESTAMP	NOT NULL, DEFAULT NOW()	
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Auto-updates on status change

Note: CHECK(customer_id <> provider_id) prevents self-booking at DB level.

reviews

Column	Type	Constraints	Notes
review_id	INT	PK, AUTO_INCREMENT	Surrogate key
booking_id	INT	FK → bookings.booking_id, UNIQUE	One review per booking enforced
reviewer_id	INT	FK → users.user_id	Must be the customer
rating	TINYINT	CHECK(rating BETWEEN 1 AND 5)	1–5 star rating
comment	TEXT	NOT NULL	
created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	

Note: UNIQUE(booking_id) ensures a booking can only be reviewed once.

conversations

Column	Type	Constraints	Notes
conversation_id	INT	PK, AUTO_INCREMENT	Surrogate key
user_a_id	INT	FK → users.user_id	Lower user_id always stored here
user_b_id	INT	FK → users.user_id	Higher user_id always stored here
created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	

Note: CHECK(user_a_id < user_b_id) + UNIQUE(user_a_id, user_b_id) prevents duplicate threads.

messages			
Column	Type	Constraints	Notes
message_id	INT	PK, AUTO_INCREMENT	Surrogate key
conversation_id	INT	FK → conversations.conversation_id, CASCADE	
sender_id	INT	FK → users.user_id	Must be user_a or user_b
content	TEXT	NOT NULL	
is_read	BOOLEAN	NOT NULL, DEFAULT FALSE	Read receipt flag
sent_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	
Note: Indexes on conversation_id, sender_id, is_read for threading and unread counts.			

reports			
Column	Type	Constraints	Notes
report_id	INT	PK, AUTO_INCREMENT	Surrogate key
reporter_id	INT	FK → users.user_id	Who filed the report
target_type	ENUM	NOT NULL	'service', 'review', 'user'
target_id	INT	NOT NULL	ID of the reported entity
reason	ENUM	NOT NULL	'spam', 'inappropriate', 'fraud', 'other'
description	TEXT	NULL	Optional detail
status	ENUM	DEFAULT 'Pending'	'Pending', 'Resolved', 'Dismissed'
reviewed_by	INT	FK → users.user_id, NULL	Admin who actioned it
created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	

ENTITY RELATIONSHIPS SUMMARY

From	Relationship	To	Rule
services.user_id	Many → One	users.user_id	CASCADE DELETE
services.category_id	Many → One	categories.category_id	RESTRICT
bookings.service_id	Many → One	services.service_id	CASCADE DELETE
bookings.customer_id	Many → One	users.user_id	RESTRICT
bookings.provider_id	Many → One	users.user_id	RESTRICT
reviews.booking_id	One → One	bookings.booking_id	CASCADE DELETE
reviewsReviewer_id	Many → One	users.user_id	RESTRICT
conversations.user_a_id	Many → One	users.user_id	RESTRICT
conversations.user_b_id	Many → One	users.user_id	RESTRICT
messages.conversation_id	Many → One	conversations.conversation_id	CASCADE DELETE

messages.sender_id	Many → One	users.user_id	RESTRICT
reports.reporter_id	Many → One	users.user_id	RESTRICT
reports.reviewed_by	Many → One	users.user_id (admin)	RESTRICT, NULL

NORMALIZATION NOTES

1NF	All columns are atomic. ENUM types used for constrained sets rather than free text.
2NF	No partial dependencies — every non-key attribute depends on the full PK (all PKs are single-column surrogates).
3NF	No transitive dependencies. categories extracted from services; conversations separated from messages; provider_id in bookings intentionally denormalized for query performance with a comment in schema.
Indexes	Strategic indexes added on all FK columns and high-cardinality filter columns (is_active, booking_status, is_read, target_type).
Constraints	CHECK constraints enforce business rules at the DB layer: price > 0, rating BETWEEN 1 AND 5, customer_id <> provider_id, user_a_id < user_b_id.